Talha Khan

Senior Game Developer - Software Engineer

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PROFILE

Unity/C# engineer owning the game lifecycle: gameplay, tools, online services, optimization, LiveOps, DevOps, testing and Android/iOS. Hands-on leadership, F2P economy/monetization and reviewed AI CLI/MCP workflows.

Portfolio: spideyatwork03.github.io/progaminfinite

EXPERIENCE

9+ years in Unity/C# development; 30+ published mobile titles. Selected public games exceed 100M combined downloads.

Ajax Studios — Lahore, Pakistan — Aug 2019 - Present

Technical Team Lead | Progressed through Junior, Senior and Dev Lead roles

Balanstein — UNITY | UGS | FIREBASE | ADS + IAP

Physics-driven F2P arcade mobile game | Active production development, unreleased

- **Built Balanstein from scratch in Unity** and own its complete engineering lifecycle through current production development.

GAMEPLAY, BOOSTERS & UPGRADES

- **Designed F2P economies and progression** with resource sinks, retention/replayability and adaptive difficulty driven by player performance and recorded collision events.
- **Implemented act/checkpoint orchestration,** scheduled obstacles and cart structures, polygon-based collection, scoring, lives and streaks.
- **Built 18 custom inspectors, six editor windows and four property drawers** for recording, assignment, baking and hierarchy automation; added snapshot continuation, stitching and thinning tools supporting approximately 800 levels.

ARCHITECTURE & PERFORMANCE

- **Engineered record-and-replay physics** with a fixed-step clock, compressed versioned binary recordings, jointed-body playback, checkpoint snapshots and recorded-velocity handoff to live physics.
- **Applied SOLID and facade/strategy/observer patterns** using injected services, immutable results and owned cancellation sources; documented 47 architecture decisions, with refactoring and code review.
- **Built animated UI** for mobile aspect ratios and 28 languages, including Arabic RTL, with Addressables; integrated VFX and character animation with technical artists.
- **Profiled CPU/GPU frame time, memory, GC and rendering** against a 30-60 FPS target on 3 GB RAM devices, using findings to guide optimization.

ONLINE SERVICES & LIVEOPS

- **Implemented Anonymous, Unity Player Accounts, Facebook and Google Play Games** authentication with account linking, stable identity, conflict handling and cached-session restoration; validated Android flows through Play internal testing.
- **Built versioned cloud persistence** with optimistic write locks, local caching, event-driven pending synchronization and lifecycle flushes; a controlled test batched 28 changes into one cloud write on flush.
- **Built and tested Local, Weekly and Global leaderboards** with shared data-source interfaces, fixed reusable rows, cached presentation and paginated UI.
- **Integrated UGS/Firebase services,** analytics, Remote Config, Firestore and FCM, plus local notifications; modeled service costs for infrastructure planning.

AJAX STUDIOS - LEADERSHIP & PRODUCTION

- **Grew engineering from one developer to eight** through hiring, training and mentoring while remaining hands-on with implementation, architecture, debugging and code quality.
- **Delivered Android/iOS builds and studio store releases** with internal, alpha and closed QA, automated Unity tests and CI/CD; introduced reviewed AI-assisted CLI/MCP workflows for implementations and tooling.
- **Implemented ad mediation, IAP and attribution** with AppLovin MAX, AdMob, Unity Ads, AppsFlyer and Adjust, connected to monetization analytics and A/B testing.

EARLIER EXPERIENCE

ZEKAB International Limited
Rawalpindi, Pakistan — May 2018 - Jun 2019

Game Developer

- **Shipped 16 Unity titles to Google Play in 13 months** from approximately 20 prototypes, building gameplay, progression, IAP, ads, leaderboards and achievements.
- **Helped establish** the studio's Unity workflow and mentored junior developers and interns.

Desire Studios
Rawalpindi, Pakistan — Jul 2017 - Apr 2018

Game Developer | Promoted from Intern

- **Designed, developed and published 18 mobile titles** across hyper-casual, simulation and puzzle genres, handling concept-to-deployment engineering and device optimization with designers and artists.
- **Awarded Employee of the Month five consecutive times** for the lowest crash rates in my games and creative ideas.
- **Refactored and optimized code for 23 additional studio games** to reduce crash rates and fatal errors.

TECHNICAL SKILLS

Unity & Gameplay
Unity 2D/3D, C#, ScriptableObjects, custom controllers, Input System/EnhancedTouch, Blend Trees, pooling, Unity Splines/Mathematics, save/load, Netcode

Software Engineering
OOP, SOLID, clean architecture, patterns, data structures/algorithms, async/await, dependency injection, cancellation, refactoring, code review

Graphics & Animation
URP/HDRP, Shader Graph, HLSL shaders, particles, post-processing, VFX, character animation, technical-artist collaboration

UI & Localization
uGUI, pixel-perfect animated UI, Canvas/Sprite Atlas optimization, aspect ratios, localization, Arabic RTL, virtualized lists

Profiling & Diagnostics
Unity Profiler, CPU/GPU/device profiling, frame time/FPS, memory, GC/draw-call reduction, Android logcat, device debugging

Testing & QA
Unity Test Framework, automated tests, internal/alpha/closed testing, cross-platform/device validation, bug tracking, pre-release QA

LiveOps & Backend
UGS Auth/Cloud Save/Cloud Code/Leaderboards/Analytics/Remote Config; Firebase Auth/Firestore/FCM/Analytics/Remote Config; REST, Unity Mobile Notifications

F2P & Game Economies
Economy design, resource sinks, progression, retention, replayability, adaptive difficulty, game business models, monetization analytics, A/B testing

Monetization & Attribution
AppLovin MAX, AdMob, Unity Ads, ad mediation, SDK integration, in-app purchases, AppsFlyer, Adjust

Release Pipeline & DevOps
Android/iOS, Play/App Store, IL2CPP/ARM64, Xcode, Android Studio, CI/CD, Unity Cloud Build, Git, Addressables, Asset Bundles, IMGUI/UI Toolkit editors

Supporting Languages & AI
C++, Python, Java, JavaScript; Claude Code, Unity MCP, Cursor, Codex; reviewed AI-assisted CLI/MCP implementation

Leadership & Production
Hiring, mentoring, technical guidance, Agile/sprint planning, code review, collaboration with designers, artists and QA

GENRES & GAME MODELS

Vehicle simulation / Physics arcade / Hyper-casual / Puzzle / Action & shooter / Free-to-play / Live service

EDUCATION

Riphah International University
BS (Hons) Software Engineering
Islamabad, Pakistan | 2013 - 2017